



NQRUST-HYPERVISOR

Your Servers | Your Cloud

A turnkey hyperconverged infrastructure platform that unifies compute, storage, and networking on commodity bare metal. KVM-based virtualization, distributed block storage, and a single web console — fully on your own premises.

All-in-1
Infrastructure

KVM
Bare Metal
VMs

HA
Multi-Node
Cluster

HCI
Compute +
Storage

Version 2.0 - Brochure
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INTRODUCTION

What Is NQRust-HyperVisor?

A turnkey hyperconverged infrastructure (HCI) platform — compute, storage, and networking unified on commodity bare-metal servers, managed from a single web console.

NQRust-HyperVisor consolidates virtualization, distributed storage, and software-defined networking into one operator-friendly platform. It runs on **commodity bare-metal hardware** — no proprietary appliances, no external SAN required — and is operated entirely on-premise for organizations that need full control over their workloads, data, and audit trail.

The platform uses **KVM hardware virtualization** for VM workloads and built-in distributed block storage that replicates volumes across cluster nodes. Every action — VM lifecycle, snapshot, backup, networking — is accessible via the web UI, REST API, or Terraform provider. Installation runs from a bootable ISO or PXE, with full air-gapped support for restricted environments.

KVM

VLAN

VXLAN

iSCSI

NFS

S3

Core Capabilities



VM Lifecycle

Create, edit, clone, and delete VMs from a web console. SSH-key injection, cloud-init configuration, and graphical & serial console access. Live migration between nodes with zero downtime.



Distributed Storage

Built-in distributed block storage replicates volumes across cluster nodes. Storage tiering, snapshot, clone, and export. VM backup to NFS or S3-compatible object storage. External iSCSI integration for existing SAN.



Software-Defined Networking

VLAN-based networking with multiple NICs per host, bonded interfaces for high availability, virtual IP (VIP) for cluster ingress, and VXLAN overlay for cross-host VM communication.

One Appliance. Four Layers.



Compute

KVM-based hypervisor for VM lifecycle, live migration, snapshots, and templates.



Storage

Distributed block storage with snapshot, clone, and backup to NFS or S3.



Network

VLAN, NIC bonding, VIP for HA, and VXLAN overlay for cross-host VM traffic.



Operator Console

Single web UI plus REST API and Terraform provider for full automation.

PLATFORM CAPABILITIES

Features That **Run** Your Workloads

From VM lifecycle to distributed storage, networking, and operator automation — built for production data centers.



VM Lifecycle Management

Create, edit, clone, and delete VMs from the web UI. SSH-key injection, cloud-init, graphical and serial console. Save VMs as reusable templates for consistent deployments.



Live Migration

Move running VMs between cluster nodes with zero downtime. Works on standard VLAN networks as well as VXLAN overlay networks for distributed deployments.



Snapshots & Restore

VM snapshots capture memory, disk, and configuration state. Restore in place or use as the base for new VMs. Storage migration is supported when restoring from snapshot or backup.



VM Backup & DR

Back up VMs to an NFS server or S3-compatible object storage. Restore failed VMs in place or create new VMs in a different cluster. Scheduled backup via Terraform provider.



Distributed Block Storage

Built-in distributed block storage replicates volumes across cluster nodes — no external SAN required. Storage tiering, online volume expansion, and export to standard formats. External iSCSI/NFS attachable.



VLAN & VXLAN Networking

VLAN trunking with multiple NICs per host. VXLAN overlay for cross-host VM networking. VPC topology view for inspecting routes and segments at a glance.



NIC Bonding & VIP

NIC bonding (active-backup default; LACP supported with MLAG-capable upstream switches) for resilient host connectivity. Virtual IP (VIP) ensures cluster ingress survives individual node failures.



REST API + Terraform

Every action available in the UI is API-accessible. Official Terraform provider for declarative provisioning of VMs, networks, volumes, and scheduled backups.



Monitoring & Logging

Built-in dashboards for cluster, host, and VM metrics. Centralized log aggregation for hypervisor and storage events. Support bundle generation for troubleshooting.

USE CASES

Built for **Every** Infrastructure**Government & Ministries**

Air-gapped deployment for classified workloads. Multi-node HA cluster on ministry-owned hardware. RBAC and audit trail for compliance. **PDN / TKDN-ready** deployment in Indonesian data jurisdiction.

**Enterprise Private Cloud**

Consolidate legacy VMware or proprietary HCI estates onto commodity bare metal. **Predictable licensing**, no per-socket fees, no surprise renewal increases. Run existing VM workloads with familiar operations.

**Edge & Remote Sites**

Small-footprint clusters at branch offices, factories, or operational sites. VXLAN overlay connects edge clusters to the central data center. Air-gapped install supports locations with intermittent connectivity.

**Dev/Test & Internal Platforms**

Self-service VM provisioning for internal engineering teams. REST API and Terraform provider integrate into existing CI/CD pipelines. Multi-user RBAC keeps environments isolated.

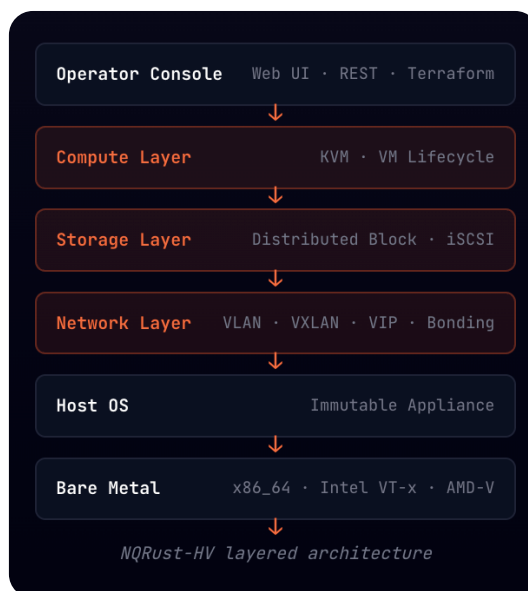
UNDER THE HOOD

Compute, storage, networking. **One appliance.**

NQRust-HyperVisor runs as an **immutable appliance** on each bare-metal host. The OS, hypervisor, storage agent, and networking control plane are delivered as one integrated unit — no separate package versions to align, no per-component lifecycle to manage.

The **compute layer** uses KVM for hardware-assisted virtualization. The **storage layer** turns local disks on each node into a **replicated distributed block store** — VMs and volumes are protected against host failure without external SAN.

The **networking layer** handles VLAN trunking, NIC bonding, and VXLAN overlay for cross-host VM communication. The **operator console** sits on top — web UI, REST API, and Terraform provider for every action.



WHY NQRUST - HYPERVISOR

More Than HCI. Infrastructure You Control.

Consolidate compute, storage, and networking on hardware you own — and operate it with predictable cost, predictable lifecycle, and complete data sovereignty.

Data Sovereignty by Design

Runs entirely on your servers. VM disks, snapshots, backups, and audit logs stay on-premise. Air-gapped capable for classified environments. Suitable for PDN / TKDN compliance and Indonesian data jurisdiction.

Predictable Annual Cost

Flat annual subscription per cluster — no per-socket or per-VM metering. No FX exposure. No surprise renewal increases. One price covers the full HCI stack and local support.

Single Pane of Glass

VMs, storage volumes, networks, hosts, and users — all from one web console. REST API and Terraform provider for automation. Dramatically reduces tool sprawl compared to legacy multi-vendor stacks.

Commodity Hardware, Integrated Stack

No proprietary appliances. Runs on standard x86_64 servers from any vendor. Compute, storage, and networking are delivered as one integrated unit — no version-matching, no separate lifecycles.

Ready to consolidate your data center?

Contact our team for a scoping discussion, sizing exercise, and installation plan. Pilot deployments and proof-of-concept options available. Quotes returned within 48 hours.

► LANDING PAGE

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► LIVE DEMO

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► DOCUMENTATION

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